Recall

27/08/2024

Para Inf

Suppose $\delta(\underset{\sim}{X})$ is a point est. of $\psi(\theta)$

$$MSE_\theta(\delta) = \mathbb{E}_\theta[\delta - \psi(\theta)]^2 = V_\theta[\delta(X)] + Bias^2(\delta, \psi(\theta))$$

$\downarrow$ one of the most popular one

$\downarrow$ only unbiasedness is not enough

If δ_1, δ_2 are u.e. of $\psi(\theta)$,
then compare δ_1, δ_2 using their variance

Def^n: If δ_1, δ_2 are u.e. of $\psi(\theta)$ with $\mathbb{E}_\theta(\delta_1^2), \mathbb{E}_\theta(\delta_2^2) < \infty$ $\forall \theta$, the relative efficiency of δ_1 w.r.t. δ_2 is

$$e_\theta(\delta_1|\delta_2) = \frac{V_\theta(\delta_2)}{V_\theta(\delta_1)}$$

δ_1 preferred if $e > 1$ $\forall \theta$

H.W $X_1, \dots, X_n \overset{iid}{\sim} U(0, \theta)$

$$\delta_1 = \frac{n+1}{n} X_{(n)}, \quad \delta_2 = 2\bar{X}$$

then $e_\theta(\delta_1|\delta_2) = \frac{n+2}{3} > 1$

Def^n: If δ_1 and δ_2 are two estimators of $\psi(\theta)$ with $\mathbb{E}_\theta(\delta_1^2) < \infty$, $\mathbb{E}_\theta(\delta_2^2) < \infty$ then relative efficiency of δ_1 w.r.t. δ_2 is

$$e_\theta(\delta_1|\delta_2) = \frac{MSE_\theta(\delta_2)}{MSE_\theta(\delta_1)}$$

δ_1 preferred if $e_\theta > 1$

Ex. $X_1, \dots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ $\qquad S^2 = \frac{1}{n-1} \sum_1^n (X_i - \bar{X})^2$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_1^n (X_i - \bar{X})^2 = \frac{n-1}{n} S^2$$

$$MSE(S^2) = \frac{2}{n-1} \sigma^4$$

$$MSE(\hat{\sigma}^2) = \frac{2n-1}{n^2} \sigma^4$$

$$e_{\sigma^2}(\hat{\sigma}^2 | S^2) = \frac{MSE(S^2)}{MSE(\hat{\sigma}^2)} = \frac{2n^2}{(n-1)(2n-1)} > 1$$

$MSE(\hat{\sigma}^2)$ is improved by "trading off variance for bias"

Loss fn. $L(\delta, \theta) \geq 0 \quad \forall\, \delta, \theta$

$L(\psi(\theta), \theta) = 0 \quad \forall\, \theta$

Risk fn. $R(\delta, \theta) = \mathbb{E}[L(\delta, \theta)]$

Ex.

1. $L(\delta, \theta) = (\delta - \psi(\theta))^2$
2. $L(\delta, \theta) = |\delta - \psi(\theta)|$
3. $L(\delta, \theta) = \mathbb{1}(\delta \neq \psi(\theta))$ [0-1 Loss]
4. $L(\delta, \theta) = \frac{\delta}{\psi(\theta)} - 1 - \ln\left(\frac{\delta}{\psi(\theta)}\right)$ [Stein's Loss]

If $\delta \to \infty$ $(\delta-\sigma^2)^2 \longrightarrow \infty$ (gross overestimation gets infinite penalty)
$\delta \to 0$ $(\delta-\sigma^2)^2 \longrightarrow 0$ (gross underestimation gets finite penalty)

If Stein's Loss used

$$\frac{\delta}{\sigma^2} - 1 - \ln\frac{\delta}{\sigma^2} \quad \begin{aligned} &\longrightarrow \infty \text{ if } \delta \to \infty \\ &\longrightarrow \infty \text{ if } \delta \to 0 \end{aligned}$$

Both gross under and overestimation is equally penalized

Consider, a class of est. $\{bs^2 : b>0\}$ and Stein's loss then s^2 gives (i.e. for $b=1$), the minimum risk (HW)

Ideally: We want to have δ^* s.t. $MSE_\theta(\delta^*) \leq MSE_\theta(\delta)$ $\forall \delta$

If $\theta=\theta_0$ take $\delta = \psi(\theta_0)$ then δ is best for $\psi(\theta_0)$

But if $\theta=\theta_1$ $\delta = \psi(\theta_1)$ is the best for $\psi(\theta_1)$

$\nexists$ uniformly best est. δ. The class of <u>all estimators</u> is for large to get the optimum. So we restrict attention to another class of u.e. of $\psi(\theta)$ and try to find the one having minimum variance!

<u>Defn</u>: δ^* is called the uniformly minimum variance unbiased est (UMVUE) of $\psi(\theta)$ if

(i) $E_\theta(\delta^*) = \psi(\theta)$ $\forall \theta$ and

(ii) $V_\theta(\delta^*) \leq V_\theta(\delta)$ $\forall \theta$ and for any u.e. δ of $\psi(\theta)$

Rigorous: Class of u.e. of $\psi(\theta)$

$$\mathcal{U} = \{\delta : \mathbb{E}_\theta(\delta) = \psi(\theta)\ \forall \theta \text{ and } \mathbb{E}_\theta(\delta^2) < \infty\ \forall \theta\}$$

δ^* is the UMVUE of $\psi(\theta)$, if $\delta^* \in \mathcal{U}$,

$$V_\theta(\delta^*) \leq V_\theta(\delta)\ \forall \theta\ \forall \delta \in \mathcal{U}$$

Two ways to find UMVUE

× (1) Cramer Rao Lower Bound

✓ (2) Lehmann-Scheffe + Rao Blackwell

Rao-Blackwell Thm:

Suppose T is a sufficient statistic for θ and δ is an unbiased estimator of $\psi(\theta)$ (i.e. $\delta \in \mathcal{U}$)

Define $\phi_\delta(T) = \mathbb{E}[\delta | T]$

(1) $\phi_\delta(T)$ is a statistic

(2) $\phi_\delta(T)$ is an unbiased estimator of $\psi(\theta)$

(3) $\mathrm{Var}\, \phi_\delta(T) \leq \mathrm{Var}\, \delta\ \ \forall \theta \in \Theta$ with "=" holds iff $\delta = \mathbb{E}[\delta|T]$ wp 1

Pf: (1) holds since T is sufficient

(2) $\mathbb{E}\phi_\delta(T) = \mathbb{E}\,\mathbb{E}(\delta|T) = \mathbb{E}\delta = \psi(\theta)$

(3) $\mathrm{Var}\,\phi_\delta(T) = \mathrm{Var}\,\mathbb{E}(\delta|T) = \underbrace{\mathrm{Var}\,\delta - \overbrace{\mathbb{E}\,\mathrm{Var}(\delta|T)}^{\geq 0}}_{\leq \mathrm{Var}\,\delta}$ ... (*)

Equality holds in ⊛ if $\mathbb{E}\,Var(\delta|T) = 0$

$Var(\delta|T) = 0$ wp 1

If $Var X = 0 \Rightarrow X = \mathbb{E}X$ wp 1

So $\delta = \mathbb{E}(\delta|T)$ wp 1

Remark: ① Given any δ from $\mathcal{U}$ we can provide an improved estimator

$$\phi_\delta(T) = \mathbb{E}(\delta|T)$$

which is a function of suff. stat

This process is called Rao-Blackwellization

② To look for UMVUEs, it it enough to consider the unbiased estimators ($\delta(T)$) that are functions of sufficient statistic

If δ is a function of T then "=" holds in ⊛

Lehmann-Schefte Thm

Suppose T is a complete sufficient statistic for θ and δ is an unbiased estimator of $\psi(\theta)$ (i.e. $\delta \in \mathcal{U}$)

Define $\phi(T) = \mathbb{E}[\delta|T]$. Then $\phi(T)$ is the UMVUE of $\psi(\theta)$

Pf: Suppose δ_1 and δ_2 are two unbiased estimators of $\psi(\theta)$ i.e. $\delta_1, \delta_2 \in \mathcal{U}$

I have $\phi_{\delta_1}(T) := \mathbb{E}(\delta_1 | T) \in \mathcal{U}$

$$\phi_{\delta_2}(T) := \mathbb{E}(\delta_2 | T) \in \mathcal{U}$$

$$\mathbb{E}[\phi_{\delta_i}(T)] = \mathbb{E}[\mathbb{E}(\delta_i | T)] = \mathbb{E}\,\delta_i = \psi(\theta) \;\forall \theta \quad i = 1, 2$$

$$\mathbb{E}[\underbrace{\phi_{\delta_1}(T) - \phi_{\delta_2}(T)}_{g(T) \text{ fn of c.s.s } T}] = \psi(\theta) - \psi(\theta) = 0 \;\forall \theta$$

$$\Rightarrow \mathbb{P}_\theta[\phi_{\delta_1}(T) = \phi_{\delta_2}(T)] = 1 \;\forall \theta$$

$$\therefore \phi_{\delta_1}(T) = \phi_{\delta_2}(T) \text{ wp } 1$$

So $\phi(T) = \mathbb{E}(\delta | T)$ does not depend on δ.

Rao-Blackwell $\Rightarrow V_\theta(\delta | T) \leq V_\theta(\delta) \;\forall \delta$

$\Rightarrow \phi(T)$ is the UMVUE of $\psi(\theta) \;\forall \theta$

Note: Suppose T_1, T_2 are c.s.s. for θ. Then $\phi(T_1)$ and $\phi(T_2)$ are both UMVUE! (argue why $\phi(T_1) = \phi(T_2)$ wp 1)

<u>Method 1</u>: Direct conditioning to get <u>UMVUE</u>

Ex. $X_1, \ldots, X_n \overset{iid}{\sim} Poi(\lambda)$. Find UMVUE of $\psi(\lambda) = e^{-\lambda}$

Find CSS for λ. $T = \sum_1^n X_i$ is c.s.s. for λ

An u.e. of $e^{-\lambda}$ is $\mathbb{1}_{(X_1=0)} = \delta(\underset{\sim}{X})$. UMVUE of $e^{-\lambda}$ is $\mathbb{E}[\delta|T]$

Now

$$\mathbb{E}[\delta | T=t] = \mathbb{P}\left[X_1 = 0 \mid \sum_1^n X_i = t\right]$$

$$= \frac{\mathbb{P}[X_1 = 0, \overbrace{\sum_2^n X_i = t}^{\to Poi(n-1)}]}{\mathbb{P}\left[\sum_1^n X_i = t\right]}$$

$$= \frac{e^{-\lambda} \dfrac{e^{-(n-1)\lambda}((n-1)\lambda)^t}{t!}}{\dfrac{e^{-n\lambda}(n\lambda)^t}{t!}} = \left(\frac{n-1}{n}\right)^t$$

$\therefore$ UMVUE of $e^{-\lambda}$ is $\left(\dfrac{n-1}{n}\right)^{\sum X_i}$

HW Ex. $X_1, \ldots, X_n \overset{iid}{\sim} Bern(\theta)$. Find UMVUE of θ^r for $r<n$, $r \in \mathbb{N}$

$\delta(X) = \mathbb{1}(X_1 \cdots X_r = 1)$

Compute $\mathbb{E}(\delta \mid \sum_1^n X_i)$ is UMVUE.